Mental Health Data Science Project Report

1️⃣ Problem Statement

Mental health issues are increasing due to lifestyle, work pressure, and social factors. However, early detection is difficult.

This project aims to:

- Analyze mental health data
- Identify important influencing factors
- Build a system to estimate mental health risk

2️⃣ Project Structure Overview

Your project is divided into multiple modules:

01_Problem_Statement

02_Dataset

03_Analysis

04_Code

05_Product

05_Report

This shows a **complete end-to-end data science pipeline**.

3️⃣ Dataset Description (02_Dataset)

You used multiple datasets:

Data.csv

Mental_Health.csv

clean_dataset.csv

Feature_Eng.csv

Data Contains:

- Demographics (Age, Gender, Marital Status)
- Lifestyle (Sleep, Diet, Activity)
- Economic factors (Income, Employment)
- Health indicators (Mental illness, Substance abuse)

4️⃣ Data Cleaning (03_Analysis → 02_cleaning.ipynb)

You handled real-world messy data:

Problems Found:

- Missing values
- Invalid values (Age = -20, 250)
- Income inconsistencies
- Mixed categories

Cleaning Steps:

Remove null values

Fix incorrect age values

Clean income format

5️⃣ Feature Engineering (03_Analysis → 03_Feature_Engineering.ipynb)

You created meaningful features.

Examples:

- Lifestyle score
- Risk indicators
- Combined health metrics

This step improves **model performance**.

6️⃣ Exploratory Data Analysis (EDA)

(03_Analysis → 01_Data_Exploration.ipynb & 04_EDA_Visual.ipynb)

Key Analysis Done:

- Age distribution
- Sleep vs mental health
- Income vs stress
- Lifestyle vs illness

Tools Used:

- Matplotlib
- Pandas
- Power BI (PowerBI.pbix)

👉 This makes your project visually strong.

7️⃣ Machine Learning Model (04_Code)

File:

Probabilities.py

Model Purpose:

Predict:

Mental Health Risk

Inputs:

- Age
- Sleep
- Income
- Activity
- Lifestyle

Output:

Low / Moderate / High Risk

8️⃣ Web Application (05_Product)

You built a real system:

app.py

templates/

Workflow:

User Input → Flask App → Model → Prediction → Output

Features:

- User input form
- Real-time prediction
- Simple interface

9️⃣ End-to-End Pipeline

Dataset

↓

Data Cleaning

↓

Feature Engineering

↓

EDA

↓

Model Training

↓

Flask App

↓

User Input

↓

Prediction Output

🔟 Conclusion

From analysis:

- Poor sleep → higher mental health risk
- Low income → higher stress
- Sedentary lifestyle → negative impact
- Family history → strong influence

This project successfully demonstrates:

✓ Data cleaning of real-world dataset
✓ Data analysis and visualization
✓ Machine learning model
✓ Working web application

Made by Rohan Mishra
BCA E1
05024402024